Problem 10.8

[Difficulty: 3]

10.8 Data from tests of a water suction pump operated at 2000 rpm with a 12-in. diameter impeller are

Flow rate, Q (cfm)	36	50	74	88	125
Total head, H (ft)	190	195	176	162	120
Power input, $\mathcal{P}$ (hp)	25	30	35	40	46

Plot the performance curves for this pump; include a curve of efficiency versus volume flow rate. Locate the best efficiency point and specify the pump rating at this point.

Given: Data on suction pump

Find: Plot of performance curves; Best effiiciency point

Solution:

Basic equations: $\eta_p = \dfrac{P_h}{P_m}$ $\quad P_h = \rho \cdot Q \cdot g \cdot H$ (Note: Software cannot render a dot!)

$\rho = 1.94$ slug/ft^3

Q (cfm)	H (ft)	$\mathcal{P}_m$ (hp)	$\mathcal{P}_h$ (hp)	η (%)
36	190	25	12.9	51.7%
50	195	30	18.4	61.5%
74	176	35	24.6	70.4%
88	162	40	27.0	67.4%
125	120	46	28.4	61.7%

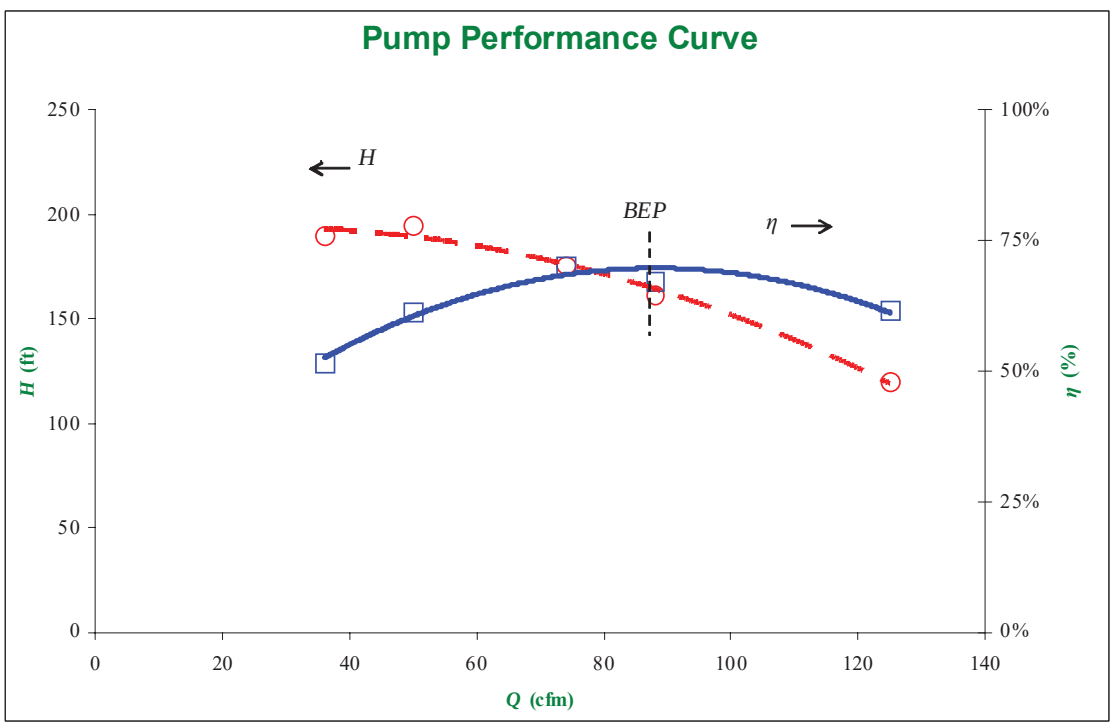
Fitting a 2nd order polynomial to each set of data we find

$H = -0.00759Q^2 + 0.390Q + 189.1$

$\eta = -6.31\text{x}10^{-5}Q^2 + 0.01113Q + 0.207$

Finally, we use Solver to maximize η by varying Q:

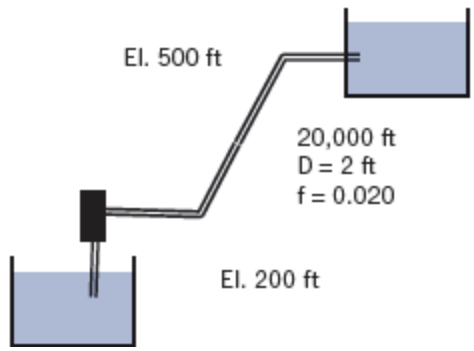
Q (cfm)	H (ft)	η (%)
88.2	164.5	69.8%



Problem 10.11

(Difficulty 2)

10.11 Using the performance curves in Appendix C, select the smallest diameter Peerless 8AE20G pump operating at 1770 rpm that will deliver a flow of at least 2000 gpm for the pipeline shown. Determine the actual flow rate and the pump electrical power requirement.



Find The smallest pump for this application.

Assumption: The flow is steady and incompressible.

Solution: Use the energy equation to determine the required head and use the pump characteristics to select a pump from Appendix C.

Applying the energy equation to the system between location 1 (the lower reservoir) and 2 (the upper reservoir) we have:

$$z_1 + \frac{p_1}{\gamma} + \frac{V_1^2}{2g} + H_p = z_2 + \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + H_l$$

Where the head loss is given by

$$h_l = f\frac{L}{D}\frac{V^2}{2g}$$

For a desired flow of 2000 gpm, the velocity in the pipe is

$$V = \frac{Q}{A} = \frac{2000\ gpm \times 0.0223\frac{ft^3}{s-gpm}}{\frac{\pi}{4}\times 2} = 1.42\frac{ft}{s}$$

The head loss is

$$h_l = 0.02 \times \frac{20{,}000ft}{2ft} \times \frac{\left(1.42\frac{ft}{s}\right)^2}{2 \times 32.2\frac{ft}{s^2}} = 6.25ft$$

For locations 1 and 2 we have

$$p_1 = p_{2,}\ V_1 = V_2 = 0,\ z_1 = 100\ ft,\ z_2 = 500\ ft$$

The energy equation becomes

$$100\ ft + H_p = 400\ ft + 6.25ft$$

The desired head is then

$$H_p = 400\ ft - 100ft + 6.25ft = 306ft$$

From Appendix C Fig. C.7 for the Peerless 8AE20G pump operating at 1770 rpm, the 18 in. diameter pump will produce a head of 320 ft at a flow rate of 2000 gpm. This is the appropriate pump to select.

The actual flow rate will be slightly larger than 2000 gpm because the head produced is larger. Selecting a large flow rate, determining the new head loss and required head, and then comparing against the pump performance curve will yield the flow rate that matches the head requirement. Through iteration, the actual flow rate will be about 2500 gpm and the total head requirement is 310 ft.

The electrical pump power is given by

$$\dot{W} = \frac{\rho Q H_p}{\eta}$$

The efficiency at 2500 gpm is 79 %. The power is then

$$\dot{W} = \frac{1.94\frac{slug}{ft^3} \times 2500gpm \times 0.0223\frac{ft^3}{s-gpm} \times 310ft \times 32.2\frac{ft}{s^2}}{0.79} = 138{,}000\ \frac{ft-lbf}{s}$$

Or

$$\dot{W} = 251\ hp$$

Problem 10.13 [Difficulty: 2]

10.13 Data from tests of a pump operated at 1450 rpm, with a 12.3-in. diameter impeller, are

Flow rate, Q (cfm)	20	40	60	80	100	120	140
Net positive suction head required, $NPSR$ (ft)	7.1	8.0	8.9	10.3	11.8	12.3	16.9

Develop and plot a curve-fit equation for $NPSHR$ versus volume flow rate in the form $NPSHR = a + bQ^2$, where a and b are constants. If the $NPSHA = 20$ ft, estimate the maximum allowable flow rate of this pump.

Given: Data on a NPSHR for a pump

Find: Curve fit; Maximum allowable flow rate

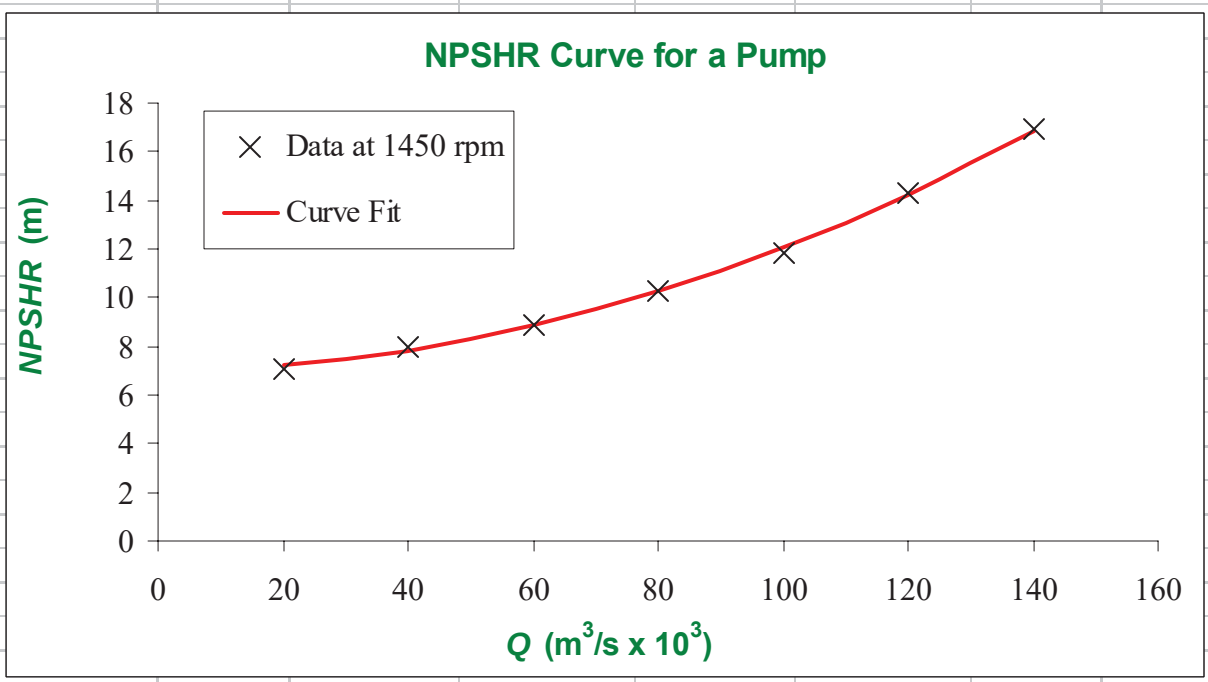
Solution: The results were generated in *Excel*:

Q (cfm)	Q^2	$NPSHR$ (ft)	$NPSHR$ (fit)
20	4.00E+02	7.1	7.2
40	1.60E+03	8.0	7.8
60	3.60E+03	8.9	8.8
80	6.40E+03	10.3	10.2
100	1.00E+04	11.8	12.0
120	1.44E+04	14.3	14.2
140	1.96E+04	16.9	16.9

The fit to data is obtained from a least squares fit to $NPSHR = a + bQ^2$

a =	7.04	ft	Q (cfm)	$NPSHR$ (ft)
b =	5.01E-04	ft/(cfm)^2	160.9	20.00

Use *Goal Seek* to find Q!



Problem 10.17

[Difficulty: 3]

10.17 Performance data for a pump are

H (ft)	179	176	165	145	119	84	43
Q (gpm)	0	500	1000	1500	2000	2500	3000

Estimate the delivery when the pump is used to move water between two open reservoirs, through 1200 ft of 12 in. commercial steel pipe containing two 90° elbows and an open gate valve, if the elevation increase is 50 ft. Determine the gate valve loss coefficient needed to reduce the volume flow rate by half.

Given: Data on pump and pipe system

Find: Delivery through system, valve position to reduce delivery by half

Solution:

Governing Equations:

For the pump and system

$$\left(\frac{p_1}{\rho} + \alpha_1 \frac{\bar{V}_1^2}{2} + gz_1\right) - \left(\frac{p_2}{\rho} + \alpha_2 \frac{\bar{V}_2^2}{2} + gz_2\right) = h_{l_T} - \Delta h_{\text{pump}}$$

where the total head loss is comprised of major and minor losses

$$h_l = f\frac{L}{D}\frac{\bar{V}^2}{2}$$

$$h_{l_m} = f\frac{L_e}{D}\frac{\bar{V}^2}{2}$$

$$h_{l_m} = K\frac{\bar{V}^2}{2}$$

Hence, applied between the two reservoir free surfaces ($p_1 = p_2 = 0,\ V_1 = V_2 = 0,\ z_1 - z_2 = \Delta z$) we have

$$g\cdot\Delta z = h_{lT} - \Delta h_{pump}$$

$$h_{lT} + g\cdot\Delta z = g\cdot H_{system} + g\cdot\Delta z = \Delta h_{pump} = g\cdot H_{pump}$$

or

$$H_{lT} + \Delta z = H_{pump}$$

where

$$H_{lT} = \left[f\cdot\left(\frac{L}{D} + 2\cdot\frac{L_e}{D_{elbow}} + \frac{L_e}{D_{valve}}\right) + K_{ent} + K_{exit}\right]\cdot\frac{V^2}{2\cdot g}$$

The calculations performed using *Excel* are shown on the next page:

Given or available data **(Note: final results will vary depending on fluid data selected)**:

L =	1200	ft	K_{ent} =	0.5	
D =	12	in	K_{exp} =	1	
e =	0.00015	ft	L_e/D_{elbow} =	30	
ν =	1.23E-05	ft^2/s	L_e/D_{valve} =	8	
Hz =	-50	ft			

The pump data is curve-fitted to $H_{pump} = H_0 - AQ^2$.
The system and pump heads are computed and plotted below.
To find the operating condition, *Solver* is used to vary Q
so that the error between the two heads is minimized.

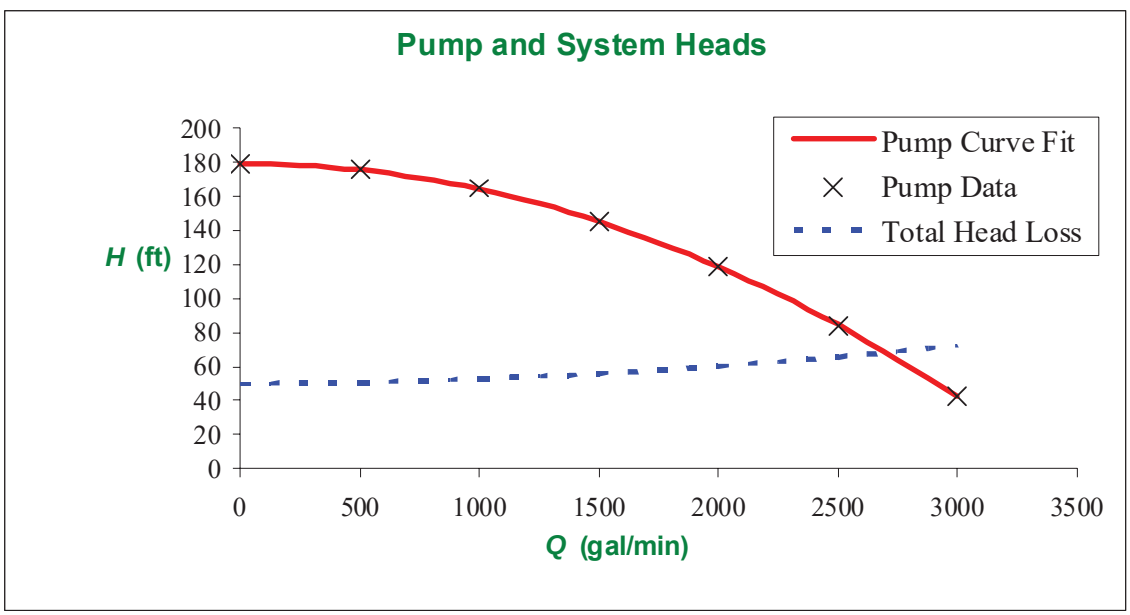
Q (gpm)	Q^2 (gpm)	H_{pump} (ft)
0	0	179
500	250000	176
1000	1000000	165
1500	2250000	145
2000	4000000	119
2500	6250000	84
3000	9000000	43

V (ft/s)	Re	f
0.00	0	0.0000
1.42	115325	0.0183
2.84	230649	0.0164
4.26	345974	0.0156
5.67	461299	0.0151
7.09	576623	0.0147
8.51	691948	0.0145

H_{pump} (fit)	H_{lT} + Hz (ft)
180	50.0
176	50.8
164	52.8
145	56.0
119	60.3
84.5	65.8
42.7	72.4

H_0 = 180 ft
A = 1.52E-05 $ft/(gpm)^2$

Q (gpm)	V (ft/s)	Re	f	H_{pump} (fit)	H_{lT} + Hz (ft)	Error)
2705	7.67	623829	0.0146	68.3	68.3	0%



For the valve setting to reduce the flow by half, use *Solver* to vary the value below to minimize the error.

L_e/D_{valve} = 26858

Q (gpm)	V (ft/s)	Re	f	H_{pump} (fit)	H_{lT} + Hz (ft)	Error)
1352	3.84	311914	0.0158	151.7	151.7	0%

Problem 10.18

(Difficulty: 3)

10.18 A fire nozzle is supplied through 300 ft of 3-in-diameter canvas hose (with $e = 0.001\ ft$). Water from a hydrant is supplied at 50 psig to a booster pump on board the pumper truck. At design operating conditions, the pressure at the nozzle inlet is 100 psig, and the pressure drop along the hose is 33 psi per 100 ft of length. Calculate the design flow rate and the maximum nozzle exit speed. Select a pump appropriate for this application, determine this efficiency at this operating conditions, and calculate the power required to drive the pump.

Assumptions: (1) $\bar{V}_1 = \bar{V}_2$ (2) $z_1 = z_2$ (3) $h_{lm} = 0$

Solution: Apply the energy equation for pipe flow with head loss:

$$\left(\frac{p_1}{\rho} + \alpha_1 \frac{\bar{V}_1^2}{2} + gz_1\right) - \left(\frac{p_2}{\rho} + \alpha_2 \frac{\bar{V}_2^2}{2} + gz_2\right) = h_{lT} = h_l + h_{lm}$$

$$h_l = f\frac{L}{D}\frac{\bar{V}^2}{2}$$

The velocity and the elevation are the same at both locations and so the energy equation reduces to

$$\frac{p_1}{\rho} - \frac{p_2}{\rho} = \frac{\Delta p}{\rho} = f\frac{L}{D}\frac{\bar{V}^2}{2}$$

$$\frac{e}{D} = \frac{0.001\ ft}{3\ in} \times 12\ \frac{in}{ft} = 0.004$$

Assuming that the flow is in the wholly rough region, from Figure 8.13 the friction factor is

$$f = 0.028$$

The average velocity is then

$$\bar{V} = \left[\frac{2}{0.028} \times 3\ in \times 3 \times (33)\ \frac{lbf}{in^2} \times \frac{ft^3}{1.94\ slug} \times \frac{1}{300\ ft} \times 12\ \frac{in}{ft} \times \frac{slug \cdot ft}{lbf \cdot s^2}\right]^{\frac{1}{2}} = 20.9\ \frac{ft}{s}$$

$$Q = \bar{V}A$$

$$A = \frac{\pi D^2}{4} = \frac{\pi}{4} \times \left(\frac{3}{12}\ ft\right)^2 = 0.0491\ ft^2$$

$$Q = \bar{V}A = 20.9\ \frac{ft}{s} \times 0.0491\ ft^2 \times 7.48\ \frac{gal}{ft^3} \times \frac{60\ s}{min} = 461\ gpm$$

Apply Bernoulli equation to the nozzle,

$$\frac{p_2}{\rho} + \frac{V_2^2}{2} + gz_2 = \frac{p_{atm}}{\rho} + \frac{V_n^2}{2} + gz_n$$

$$V_n = \left[\frac{2p_2}{\rho} + V_2^2\right]^{\frac{1}{2}}$$

$$V_n = \left[2 \times 100\ \frac{lbf}{in^2} \times \frac{ft^3}{1.94\ slug} \times 144\ \frac{in^2}{ft^2} + \left(20.9\ \frac{ft}{s}\right)^2\right]^{\frac{1}{2}} = 124\ \frac{ft}{s}$$

The pump head requirement (neglecting the velocity head and elevation difference) will be:

$$H_p = \frac{p_1 - p_0}{\rho g} = [100 + 3 \times (33) - 50]\ \frac{lbf}{in^2} \times \frac{ft^3}{62.4\ lbf} \times 144\ \frac{in^2}{ft^2} = 344\ ft$$

From the pump selector chart (Fig. c.1) choose 3AE98 or 4AE10 pump, 3500 rpm. Based on 4AE12 at 3550 rpm, expect $\eta = 0.75$. The power is then

$$P = \frac{Q\Delta p}{\eta} = \frac{\rho g \Delta h Q}{\eta} = \frac{1}{0.75} \times 20.9\ \frac{ft}{s} \times 0.0491\ ft^2 \times 62.4\ \frac{lbf}{ft^3} \times 344\ ft \times \frac{hp \cdot s}{550\ ft \cdot lbf} = 53.4\ hp$$